

# Statement of Research Interests

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My research examines how atomic structure controls the electronic properties of two-dimensional materials. During my PhD at the University of Sydney, I have studied graphene-based heterostructures, boron allotropes, and hydrogen-functionalized beryllium sheets through density functional theory (DFT). I now wish to extend this work to twisted multilayers, using machine-learning models to investigate relaxation and electronic structure at scales beyond direct DFT.

My work on graphene–borophene and graphene–boron-carbide bilayers connects stacking registry and interlayer separation to binding, charge redistribution, and optical response. Using VASP with PBE and HSE06, I compared competing structures and examined their electronic bands and dielectric functions. Subsequent projects extended this work to phonons, molecular dynamics, magnetism, and band topology. Two studies have led to first-author publications in *Nanomaterials* and *Physical Review Materials*; the beryllium manuscript is in preparation. I have also developed reusable analysis scripts and Jupyter workflows for convergence tests and physical observables.

My machine-learning experience currently consists of small toy-model experiments in JAX. I would develop this experience further in Professor Ortner’s group while contributing my stronger background in first-principles calculations, data analysis, and physical validation.

At UBC, I would begin with twisted graphene, building on the group’s atomic cluster expansion work. I am particularly interested in how accurately an interatomic potential must describe local stacking and interlayer forces to predict moiré relaxation. I would generate reference calculations spanning registry, layer separation, strain, and selected defect environments, then test the model against relative energies, forces, and relaxed structures. These comparisons would guide additional sampling where predictions become unreliable.

Next, I would connect the relaxed geometries to a separately validated machine-learning tight-binding model, drawing on the group’s equivariant Hamiltonian methods. Comparisons with DFT bands, orbital character, and densities of states would establish its reliability before extending it to larger cells. The physical question is how corrugation, strain, and defects modify electronic localization and low-energy bands. Tracing errors from geometry to these observables would connect model development to experimentally relevant predictions. Extensions to BN or transition-metal dichalcogenides would follow the collaboration’s priorities.

My longer-term interests include many-body phenomena in two dimensions. Reliable low-energy Hamiltonians would provide a basis for exploring these questions, while my immediate focus would remain the structural and electronic surrogate models central to this position.